

WP-11 · Formalism: the interleaved bilateral construction and its two properties

WP-11 (rpp-adversarial-paper) · tier A (formal sections for the tier-A paper) · 2026-07-18
· status DRAFT v1 — definitions + one conditional reduction; every st

The Privacy-is-Value Research Programme

2026-07-18

This section formalises the two properties the RPP construction claims under the First-Person ruling (Option C): a *behavioral* property of the injection mechanism (forcing), and an *information-theoretic* property of the two-party interleaved pair (withheld-fraction security). The central discipline is that these are separate: forcing gets the proverb authored; the entropy split is what makes the result secure. Conflating them — presenting the behavioral property as a security guarantee — is the failure mode this section exists to prevent (the modal error of E6-C06: a mechanism that compels is not thereby a mechanism that cannot be broken).

1. The construction

Two parties A and B share an experience E and each author a natural-language compression P_A, P_B (their proverbs). An **interleave** map I combines them into a single object I(P_A, P_B); a public **anchor** discloses a σ -fraction of that object, $A_\sigma := \text{disc}_\sigma(I(P_A, P_B))$ for a disclosure ratio $\sigma \in [0,1]$, while the complementary **withheld fraction** $W := I(P_A, P_B) \setminus A_\sigma$ (the $(1-\sigma)$ -fraction) is held jointly by the two parties. The commitment is $\kappa := H(I(P_A, P_B))$ for a collision-resistant hash H. Verification/recovery requires reconstructing I(P_A, P_B) to within a fuzzy tolerance sufficient to recompute κ (equivalently, to open the commitment under an error-correcting opening; the hard-crypto layer — H, and the signatures/ κ -addressing of the deployed system — is assumed from the implementation record E11 and is not re-derived here).

Deployment status (GR-8, E3-C32): I and the σ -budget are design-layer objects (the deployed system uses the asymmetric-default inscription and a stratum-derived visibility, not $\sigma \in [0,1]$); this section formalises the specified construction and states this boundary.

2. Forcing (the behavioral property of the injected directive) — Definition, not a guarantee

Let a reader R (a human or a model) process content C, and let $\Phi(C)$ denote the event that R emits a proverb *bound to C* before producing any downstream answer. Let D be the injected RPP directive. Let $\mathcal{C}$ be a content distribution and $\mathcal{R}$ a reader population.

Definition 1 (forcing). The directive D is **(f, b, s)-forcing** over $(\mathcal{C}, \mathcal{R})$ if 1. (*compulsion*) $E_{\{C \sim \mathcal{C}, R \sim \mathcal{R}\}}[\Pr(\Phi(C+D)) - \Pr(\Phi(C))] \geq f$ (the directive raises the probability of the bound act by at least f

over the no-directive baseline); 2. (*content-binding*) $\Pr(\text{the emitted proverb is specific to } C \mid \Phi(C+D)) \geq b$, where “specific to C ” is failure of a fixed generic-completion test (a proverb that a reader who never saw C could also have produced does not count); 3. (*strip-survival*) under each of a declared family of pre-processing attacks $\mathcal{A}$ (directive deletion, paraphrase, relocation), the compulsion gap remains $\geq s$.

Remark 1 (what Definition 1 is and is not). Forcing is a statistical property of a reader population under a directive. It is NOT unforgeability and NOT a security guarantee: a reader that is forced to author a bound proverb has done exactly what the mechanism intends, and a capable model performing the act is a success of the mechanism, not a break of it. The security of the construction does not rest on Definition 1 at all (it rests on §3). The only role of forcing is to ensure the proverb is authored under the directive rather than skipped.

Remark 2 (forcing is the dual of injection-resistance, hence model-relative). Prompt-injection attacks seek to make a model follow an injected instruction; prompt-injection defenses seek resistance (Greshake et al. arXiv:2302.12173; OWASP LLM01). RPP’s forcing is cooperative injection: it wants the model to follow. Therefore f is inversely related to the reader’s injection-resistance, and the honest claim is bounded — forcing holds over compliant reader populations and weakens on hardened ones. The strongest positive form of the claim is that D is (f, b, s) -forcing even over an injection-resistant population (the benchmark’s arm A3, L164); the weakest admissible form scopes $\mathcal{R}$ to cooperative readers and says so. Definition 1 carries $\mathcal{R}$ explicitly so the scope is never hidden.

Distinction from neighbours, in the definition’s terms. Zero-shot chain-of-thought (Kojima et al. arXiv:2205.11916) achieves compulsion of a reasoning act (f high) with content-binding $b \approx 0$ (the reasoning is not a C -specific authored artifact bound to a commitment) and is self-directed (the operator steers their own model, not a directive embedded in third-party content). “Context bombs” (Tracebit 2026) raise a *derailment* event, not Φ , over a hostile reader. RPP is the (high f , high b , embedded-and-cooperative) corner; b is the coordinate that separates it from CoT, and the cooperative-embedded setting separates it from context bombs.

3. Withheld-fraction security (the information-theoretic property of the pair)

Fix the malicious-counterparty threat model (the honest adversary, L163): the adversary is party B , who was present at E and now holds their own proverb P_B , the shared-experience context E , and the public anchor A_σ . Write the adversary’s side information $S := (P_B, E, A_\sigma)$. The adversary wins by producing W' within fuzzy tolerance t of the withheld fraction W (enough to recompute κ). We use fuzzy conditional min-entropy in the sense of the fuzzy-extractor line (Dodis, Reyzin & Smith, EUROCRYPT 2004; Juels & Wattenberg, CCS 1999): $\tilde{H}_{\{t, \infty\}}(X \mid Z) := -\log E\{z\}[\max_{\{x'\}} \Pr(d(X, x') \leq t \mid Z = z)]$, the log of the best t -ball guessing probability given Z .

Definition 2 (interleave splitting). The interleave I is **(σ, δ) -splitting** if the σ -disclosure leaks at most δ bits about the withheld fraction: $I(A_\sigma; W) \leq \delta$. ($\delta = 0$ is an ideal interleave whose public fraction is independent of the withheld fraction; a poor interleave, e.g. a prefix cut of a redundant encoding, has large δ and is insecure regardless of proverb entropy. δ is a property of the map I , measurable directly.)

Definition 3 (informed-counterparty security). The construction is **(t, ϵ) -secure** against the malicious counterparty if the guessing probability of the withheld fraction within tolerance t given the adversary’s side information is at most ϵ : $\Pr[A(S) \text{ outputs } W' \text{ with } d(W', W) \leq t] \leq \epsilon$, equivalently $\tilde{H}_{\{t, \infty\}}(W \mid S) \geq \log(1/\epsilon)$.

Proposition 4 (security reduction — Proven-conditional). If I is (σ, δ) -splitting (Definition 2) and party A 's proverb retains fuzzy conditional min-entropy $\tilde{H}_{\{t, \infty\}}(P_A | P_B, E) \geq m$ against a co-participant who holds P_B and E , then the construction is (t, ε) -secure with $\varepsilon \leq 2^{-(m - \delta)}$. Proof. The withheld fraction W is a deterministic function of (P_A, P_B) via I . The adversary's guess of W within tolerance t is at least as hard as guessing the part of $I(P_A, P_B)$ not determined by (P_B, E) and not disclosed by A_σ . Conditioning reduces min-entropy by at most the information the conditioning carries: $\tilde{H}_{\{t, \infty\}}(W | P_B, E, A_\sigma) \geq \tilde{H}_{\{t, \infty\}}(W | P_B, E) - I(A_\sigma; W | P_B, E) \geq (m) - \delta$, where the first inequality is the standard chain bound for conditional min-entropy under side information of at most $I(A_\sigma; W | P_B, E) \leq \delta$ bits (Definition 2, taken conditionally), and the second uses that the residual entropy of W given (P_B, E) is lower-bounded by that of P_A given (P_B, E) up to the fuzzy tolerance, since W includes the $(1 - \sigma)$ -fraction of A 's contribution. The guessing bound $\tilde{H}_{\{t, \infty\}}(W | S) \geq m - \delta$ gives $\varepsilon \leq 2^{-(m - \delta)}$. ■ **Named gaps (A3 discipline — do not hand-wave).** Two steps are stated as requirements the construction must meet, not as facts: (i) the claim that W 's residual entropy given (P_B, E) is lower-bounded by P_A 's requires that the interleave place at least the withheld share of A 's independent contribution into W — an **interleave-faithfulness requirement** on I , listed as an engineering requirement, not assumed; (ii) the fuzzy-tolerance accounting (that error-correcting to within t does not itself leak more than the secure-sketch entropy loss) is the standard fuzzy-extractor caveat and is inherited, with its known entropy-loss term folded into m (i.e. m is the *post-sketch* residual entropy). Both are inherited-or-required, and both are measurable.

Corollary 5 (the estimator link). The benchmark quantity $r(\sigma)$ (L164, H2) is an empirical estimator of $\varepsilon(\sigma)$. The pre-registered falsification $r(0.618) \geq 0.5$ is exactly the formal statement $m - \delta \leq 1$ bit at $\sigma = 0.618$: the withheld majority-public sweet-spot retains at most one bit of residual entropy against an informed counterparty, i.e. the construction is insecure there. Thus the experiment measures the two quantities Proposition 4 reduces security to (δ from I , m from proverb entropy given a co-participant), and refutes or supports the reduction's premises directly.

Proposition 6 (monotonicity in σ — Conjecture, proof sketch + missing step). If I is **graceful** — each fragment moved from the disclosed set A_σ to the withheld set W as σ decreases adds non-negative conditional min-entropy to W given S — then $\varepsilon(\sigma)$ is non-increasing as σ decreases, so $r(\sigma)$ is non-increasing in σ^{-1} (more withheld $\Rightarrow$ harder). *Proof sketch.* Decreasing σ enlarges W by fragments previously in A_σ ; under gracefulness each such fragment weakly increases $\tilde{H}_{\{t, \infty\}}(W | S)$; the guessing bound of Proposition 4 then gives non-increasing ε . *Missing step (named).* Gracefulness is not automatic: an interleave can move a fragment into W that is fully determined by fragments remaining in A_σ (or by P_B), adding zero entropy, or worse, changing the correlation structure so δ rises. Whether a concrete I is graceful is a property of I to be **constructed and proven**, not assumed; absent that, monotonicity is a conjecture the benchmark's isotonic test (L164) can refute empirically. This is labelled Conjecture, not Proposition-with-■.

4. The layering (the honesty discipline this section enforces)

The two properties occupy different registers and must be stated as such throughout the paper: - **Forcing (Definition 1)** is behavioral, model-relative, and provides no security. Its role is to author the proverb under the directive. It is measured by H1. - **Security (Definitions 2–3, Proposition 4)** is information-theoretic and holds regardless of whether any reader “understood”: it rests on the residual entropy of A 's proverb given a co-participant's full knowledge, and on the interleave's splitting. It is conditional on two measurable quantities (m, δ) and on two named engineering requirements (interleave-faithfulness, gracefulness). It is measured by H2.

Consequence for the brain-wallet discharge (making L163 precise). The brain-wallet critique is the statement that a single human-memorable secret has low min-entropy against a public target. Proposition 4 shows the security here is not the min-entropy of one proverb but $\tilde{H}_{\{t,\infty\}}(P_A | P_B, E)$ — the residual entropy of A's proverb *given a co-participant's complete knowledge of the shared experience and their own half*. The critique is therefore not refuted by assertion but relocated to a measurable quantity: if a co-participant can guess A's proverb (m small), the construction is insecure, and H_2 measures exactly this. The honest threat is the informed co-participant, and the security claim is precisely " $m - \delta$ is large at the operating σ ," neither more nor less.

Coherence with the programme (cited, not conflated). Definition 3 and Proposition 4 are the same information-theoretic family as the companion's informed-adversary corollary (WP-07 Corollary 5.4b: an error floor governed by $H(X | B_t)$ against an adversary holding accumulating side information). Both bound reconstruction by a residual conditional entropy against an informed adversary. They are not the same theorem — WP-07's B_t is accumulating linkage over calendar time about a fixed archive; here S is a fixed co-participant's knowledge of a shared experience — and the paper cites the kinship without asserting identity (GR-1/GR-4 discipline: same family, distinct objects).

5. Firewall — what is proved, defined, assumed, empirical, conjectural

- **Proved-conditional:** Proposition 4 (the reduction $\varepsilon \leq 2^{-(m-\delta)}$), by standard conditional-min-entropy bounds, conditional on Definitions 2 and the two named requirements.
- **Definitions:** Definition 1 (forcing), Definitions 2–3 (splitting, security). Forcing is a definition whose *satisfaction* is empirical.
- **Empirical (measured, not assumed):** m (residual proverb entropy given a co-participant — H_2), δ (interleave leakage — direct measurement), $f/b/s$ (forcing — H_1), the σ -frontier crossover ($H_2 \times H_3$).
- **Engineering requirements (must be met by the construction, listed not assumed):** interleave-faithfulness (Prop 4 gap i), gracefulness (Prop 6), the fuzzy-sketch entropy-loss accounting (Prop 4 gap ii).
- **Conjecture:** Proposition 6 (monotonicity in σ), with the missing step named (gracefulness of a concrete I).
- **Cited substrate, not claimed:** the fuzzy-extractor/secure-sketch primitives (Dodis-Reyzin-Smith 2004; Juels-Wattenberg 1999) — RPP builds on them for the reproduce-from-noisy-input layer and does not claim that primitive (WP-11a).
- **Conjectural parameters:** the φ candidate values for σ (register C54) — candidate operating points, never asserted optima.
- **Assumed from elsewhere:** the hard-crypto layer (H, κ , signatures) from E11; not re-derived.

Nothing in this section asserts the construction unconditionally secure. The strongest statement is: *the construction is $(t, 2^{-(m-\delta)})$ -secure against an informed co-participant, where m and δ are quantities the pre-registered experiment measures, under two named interleave requirements; whether $m - \delta$ is large enough at an operating σ is exactly what H_2 decides.*

6. The concrete interleave I^* (First-Person ruling, 2026-07-18) — the construction that binds

The abstract interleave I is now fixed to a concrete construction, so Propositions 4 and 6 have an object to bind to. The choice was between a graded interleave (σ a smooth disclosure knob) and an all-or-nothing transform (maximal security but σ binary); the First Person ruled the graded path, which matches the “interleaved visibility ratio in this knowledge sharing” intent (L163) and keeps σ a real tunable knob.

Definition 7 (the concrete interleave I^* , extract-then-block-interleave). Fix a fuzzy extractor (secure sketch SS + strong extractor Ext) with public helper data and error tolerance t (Dodis-Reyzin-Smith, EUROCRYPT 2004). Then: 1. **Extract.** $s_A := Ext(SS(P_A))$, $s_B := Ext(SS(P_B))$; each is (near-)uniform of length equal to its post-sketch residual min-entropy; helpers are public. This step does two jobs: it tolerates rewording (the fuzzy part) and it whitens natural-language redundancy so blocks become (near-)independent and uniformly informative. 2. **Block-interleave.** Partition s_A , s_B into n blocks each and interleave on a fixed public symmetric schedule π (e.g. $s_A[1] s_B[1] s_A[2] s_B[2] \dots$), giving $Z := \pi(s_A, s_B)$. 3. **Disclose.** The anchor A_σ is a σ -fraction of Z ’s blocks under a public schedule that draws the same fraction from each party’s blocks; W is the complement; $\kappa := H(Z)$.

What I^* discharges (the two named gaps of §3, reduced to the extractor — not assumed). - *Splitting (Definition 2), δ bounded not merely measured.* Ext output is within statistical distance δ_{ext} of uniform, and $s_A \perp s_B$ by independent authoring (the $N2$ property). A co-participant B re-derives the s_B blocks from P_B and the public helper, so the s_B blocks in A_σ leak nothing new to B ; the s_A blocks are (near-)uniform, so disclosing a σ -fraction of them leaks at most δ_{ext} about the withheld $(1-\sigma)$. Hence $I(A_\sigma ; W \mid P_B, E) \leq \delta_{ext}$, and δ is a bounded extractor parameter, not an empirical unknown. - *Interleave-faithfulness (Proposition 4, gap i), DISCHARGED.* The symmetric schedule places $(1-\sigma)$ of s_A ’s blocks into W by construction; since s_A carries A ’s post-sketch residual entropy against B , W ’s residual entropy given the adversary’s side information is at least $(1-\sigma) \cdot m$ up to δ_{ext} , where $m := \tilde{H}\{t, \infty\}(P_A \mid P_B, E)$ post-sketch. The requirement is met by the schedule, not assumed. - *Gracefulness (Proposition 6), PROVEN for I^* .* Each block moved from A_σ to W as σ decreases is a (near-)uniform block (near-)independent of the disclosed set, so it adds (near-)its full entropy to $\tilde{H}\{t, \infty\}(W \mid S)$. Monotonicity therefore holds for I^* — Proposition 6, a Conjecture for arbitrary I , is a Proposition for I^* : $\epsilon(\sigma)$ is non-increasing as σ decreases, strictly under the uniformity of Ext .

Theorem 8 (security of I^* , Proven-conditional on the extractor and on m). Under Definition 7, the construction is (t, ϵ) -secure against the malicious informed counterparty with $\epsilon \leq 2^{-\{-(1-\sigma) \cdot m - \delta_{ext}\}}$, $m := \tilde{H}\{t, \infty\}(P_A \mid P_B, E)$ post-sketch, δ_{ext} the extractor’s uniformity loss. Proof. Proposition 4 with the I^* -discharged inputs: faithfulness gives $\tilde{H}\{t, \infty\}(W \mid S) \geq (1-\sigma) \cdot m - \delta_{ext}$; the guessing bound gives $\epsilon \leq 2^{-\{-(1-\sigma) \cdot m - \delta_{ext}\}}$. Monotonicity is Proposition 6 for I^* . ■ **The single remaining empirical unknown.** Choosing I^* collapses the two engineering requirements and the δ -measurement into the extractor’s standard guarantee (δ_{ext} bounded, faithfulness by schedule, gracefulness proven). Exactly one quantity remains empirical: **m , the residual entropy of a proverb given a co-participant who was present at the shared experience.** The entire security question is now “how guessable is A ’s proverb to someone who was there,” which is the honest threat and is exactly what benchmark $H2$ measures. A low- m proverb is insecure under any interleave; I^* guarantees you retain $(1-\sigma) \cdot m$ of whatever m is, no less.

The two rejected alternatives, recorded. - *Naive text interleave (no extractor):* interweave the raw

proverb characters/words. Human-readable, but natural-language redundancy makes disclosed fragments predict withheld ones (δ large), so security is weak and not cleanly analysable. Rejected as the security layer; it may serve only as the readable surface (below). - *All-or-nothing transform (AONT)*: withhold any fragment $\Rightarrow$ the whole is unrecoverable ($\delta = 0$, maximal security), but σ collapses to binary and the graded knowledge-sharing knob is lost. Rejected for the graded- σ intent; it remains the option if maximal security is wanted and graded disclosure is not.

Layering (keeps M2 honest). I^* whitens proverbs to bits for the SECURITY layer (Z); the human-readable proverb is a SEPARATE surface layer for knowledge-sharing and relational recall (M2/M3). σ applies to both in lockstep: the readable anchor discloses σ of the joint human-readable understanding, and the withheld $(1-\sigma)$ of Z is what protects. The two are coupled — sharing more readable knowledge (higher σ) genuinely lowers security by lowering $(1-\sigma) \cdot m$ — which is the σ -frontier the benchmark measures, not a cosmetic knob. The secure-sketch helper data's own leakage is folded into m being the post-sketch residual (the standard fuzzy-extractor accounting, inherited from the cited primitive).

Design rationale — why the graded path (the understanding-as-key thesis, restated register-neutrally). The choice of I^* over the all-or-nothing transform is not merely a usability preference: in the understanding-as-key research the disclosure-security trade IS the intended generative mechanism, framed there as *identity as a dance, not a stance* (First-Person framing). Register-neutrally: a stance is a static held secret verified by equality; a dance is an ongoing relational disclosure process in which parties continuously trade what they reveal against what they hold, and the trajectory of that trade over time — not any fixed value — is the identity. The graded σ of I^* is the formal shadow of that thesis: it makes disclosure a continuous, re-negotiable position (a dance along σ), where AONT would force a static all-or-nothing secret (a stance). The residual-entropy kernel is already present in the understanding-as-key source: “attackers cannot enumerate without understanding the relationship itself” (understanding-as-key :872) is exactly $m = \tilde{H}_{\{t,\infty\}}(P_A \mid P_B, E)$ — security as the residual uncertainty of the relationship given the adversary's own knowledge of it. GR-4/GR-8 provenance (located 2026-07-18): the phrasing “identity is the dance, not the stance” is in the corpus at poems/tide-orbit-selene.md:26 (with its development at :101-108, “the dance is the identity”), which is NARRATIVE-CANON (the seventh-capital / C55 home, per SOURCES seventh-capital-canon). Per GR-4 narrative-canon is restated register-neutrally and NOT cited directly in a tier-A artifact; the citable formal anchor is the understanding-as-key/Zypher kernel at :872 (m = residual relationship-entropy). The Zypher lineage paper carries the kernel but not the phrase; the phrase's home is the poem. (The First Person recalls a Nov-2025 understanding-as-key document also carrying it; if that is a source distinct from the poem it is a minor SOURCES-completeness item for A1, but the concept is now grounded and the formal statements stand independently of the framing regardless.)

7. Handoff

- **A6:** the definitions here fix what the metrics estimate — $r(\sigma) \leftrightarrow \varepsilon(\sigma)$ (Corollary 5), the forcing triple $\leftrightarrow (f, b, s)$ (Definition 1). No benchmark change required; the content-binding detector operationalises “specific to C” (Definition 1.2), and the interleave-leakage measurement operationalises δ (Definition 2). Add one measurement: $\delta(\sigma)$ directly ($I(A_\sigma; W)$), so Proposition 4's two inputs are both measured, not one inferred.
- **A5-pc (post-results):** the two places a PC breaks this are the named gaps in Proposition 4 (interleave-faithfulness) and the gracefulness conjecture (Proposition 6). Both are labelled and both are empirically refutable, which is the defensible posture; the review should check

that the paper never states Proposition 6 as proved.

- **A4:** cite Dodis-Reyzin-Smith 2004 and Juels-Wattenberg 1999 at primary records (both already in the WP-11a spine); confirm the fuzzy conditional min-entropy notation matches the cited source before tier-A use.
- **First Person:** the interleave I is currently abstract; a concrete I must be chosen and shown faithful+graceful for Propositions 4 and 6 to bind. This is a construction decision (which interleave), separate from the earlier architecture ruling. Until an I is fixed, the security is a template, not a theorem.